ITI 1100

Chapter 1 - Binary systems

Different numbering systems

- Decimal (Arabic): (0,1,2,3,4,5,6,7,8,9)
- Octal: (0,1,2,3,4,5,6,7)
- Hexadecimal: (0,1,2,3,4,6,7,8,9,A,B,C,D,E,F)
- Binary: (0,1)

1 - Positional Notation

$N = (a_{n-1} a_{n-2} \cdots a_1 a_0 . a_{-1} a_{-2} \cdots a_{-m})_r$

2 - Polynomial Notation

$N = a_{n-1} \times r^{n-1} + \cdots a_0 \times r^0 + a_{-1} \times r^{-1} + \cdots + a_{-m} \times r^{-m}$

$$N = \sum_{i=-m}^{n-1} a_i r^i$$

Conversions

Decimal -> Anything

Integer part
-> keep dividing by the (of the anything) base and collect the remainders in reverse order

fraction part
keep multiplying by base until it becomes "0". Collect integer parts (forward order).

Base 2^x -> Binary

convert each digit into its x number of bits binary equivalent.

Binary -> Base 2^x

split the number into groups of x and convert each group into their equivalent in Base 2^x

Anything -> Decimal: Sum of polynomial notation

Binary addition

$$\begin{array}{r} 1010\ 0011\ 1011\ 0001 \\ +\ 0111\ 0100\ 0001\ 1001 \\ \hline 1\ 0\ 001\ 0111\ 0100\ 1010 \end{array}$$

Binary subtraction

$$\begin{array}{r} 0 \\ -\ 0 \\ \hline 1 \end{array} \quad \begin{array}{r} 0 \\ -\ 1 \\ \hline 11 \end{array} \quad \begin{array}{r} 1 \\ -\ 0 \\ \hline 1 \end{array} \quad \begin{array}{r} 1 \\ -\ 1 \\ \hline 0 \end{array} \quad \begin{array}{r} 101011 \\ -\ 010101 \\ \hline 010110 \end{array}$$

Binary codes: BCD

- used to represent decimal digits 0-9
- 4 bits are used

Ex:

$(9750)_{10} =$

$(1001\ 0111\ 0101\ 000)_{BCD}$

Binary codes: ASCII

Binary multiplication

$$\begin{array}{r} 0 \\ \times\ 0 \\ \hline 0 \end{array} \quad \begin{array}{r} 0 \\ \times\ 1 \\ \hline 0 \end{array} \quad \begin{array}{r} 1 \\ \times\ 0 \\ \hline 0 \end{array} \quad \begin{array}{r} 1 \\ \times\ 1 \\ \hline 0 \end{array}$$

Binary division

$$\begin{array}{r} 0 \\ \div\ 1 \\ \hline 0 \end{array} \quad \begin{array}{r} 1 \\ \div\ 1 \\ \hline 1 \end{array}$$

Radix Complement (r's complement)

$$[N]_r = r^n - (N)_r$$

Diminished radix complement

$$[N]_{r-1} = (r^n - 1)_r - (N)_r$$

then replace 0's by 1's & 1's by 0's

Substraction with 2's complement

$$A - B = A + (-B)$$

(add 2's complement of B to A)

x: A=1010100 B=1000011
A-B = A+(-B) = A+[C

1010100
0111101
10010001
scard
nd
carry

Subtraction with 1's complement

$$A - B = A + [B]$$

(add 1's complement of B to A)

Ex: A=1010100 B=1000011
A-B = A+[B] = (1010100) + (0111100) = (0010001)
1010100 + 0111100 = 0010000
+1 = 0010001

Signed complement

a) Signed magnitude representation

1	0	0	0	0	0	1	1

-3 = 10000011 = 1/(Sign bit) 0000011

b) Signed 1's representation

1	1	1	1	1	1	0	0

-3 = 10000011 = 1/(Sign bit) 1111100

c) Signed 2's representation

1	1	1	1	1	1	0	1

-3 = 10000011 = 1/(Sign bit) 1111101

Overflow

* happens when the sum of the outside the range of the register

in signed complement overflow
-> addition of two ⊕ #'s results in ⊖
-> addition of two ⊖ #'s results in ⊕